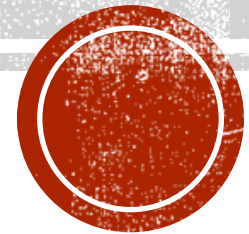


NEOSTATS DOCUMENT INTELLIGENCE

**AI-Powered Financial Document Extraction,
Validation & API Platform**



PROBLEM & OBJECTIVE

Problem

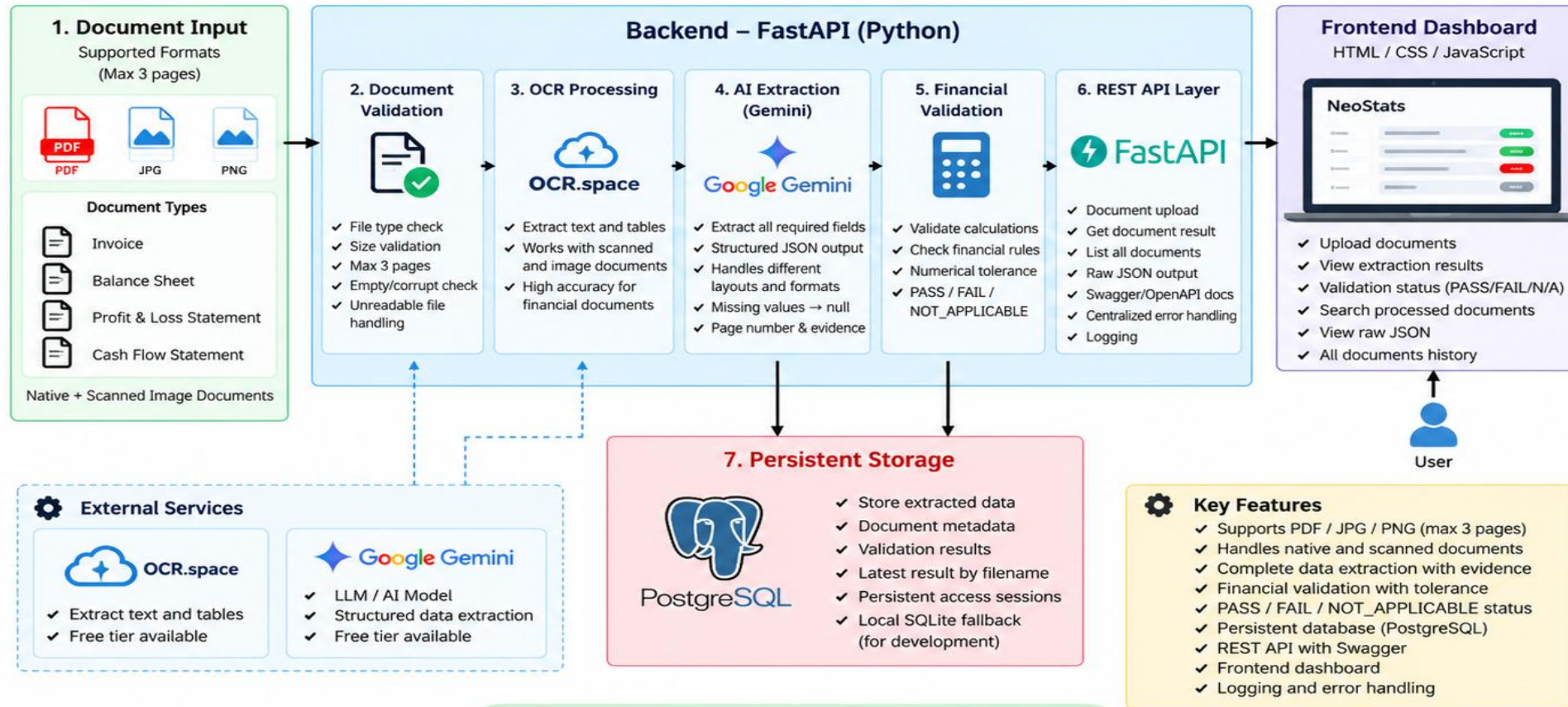
- Financial documents are often received as PDFs, scans, or images.
- Manual extraction is time-consuming and error-prone.
- Different financial statements require different validation rules.

Objective

- Automatically extract meaningful financial information.
- Preserve source evidence and page references.
- Validate extracted financial values using accounting relationships.
- Provide results through a searchable web dashboard and REST API.



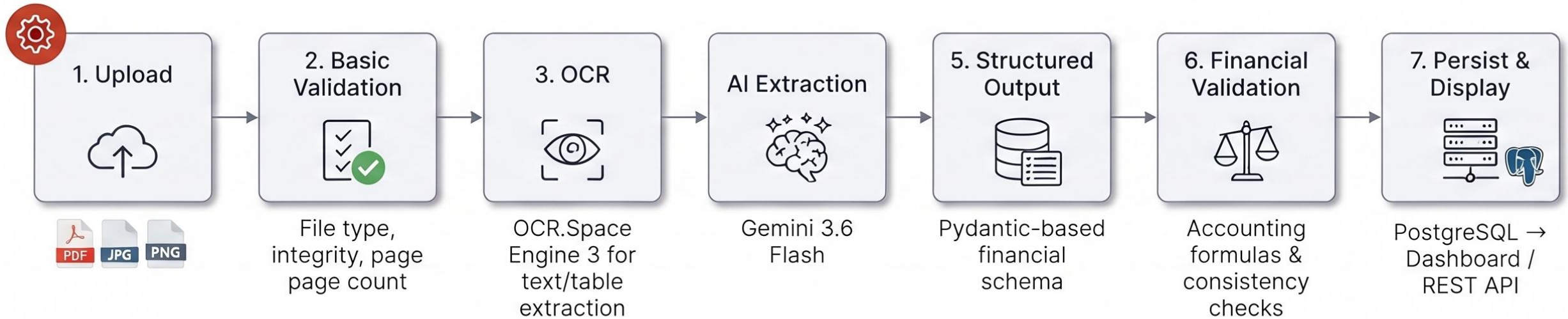
SYSTEM ARCHITECTURE



From Documents to Insights – Powered by AI
"Automating financial document analysis for a smarter tomorrow"



PROCESSING PIPELINE



EXTRACTION & GROUNDING

Extracted Information

- Document type
- Dates / reporting periods
- Units of measurement
- Financial sections
- Line items
- Values and comparative periods
- Notes
- Signatures & attestation
- Document footer



EXTRACTION & GROUNDING

Grounding

- Original document is provided to Gemini.
- OCR output is used as supplementary text/table evidence.
- Extracted values include source evidence where available.
- Page number is preserved where available.

Missing Information

- `null` → field/value is not present
- No hallucination / inference of unsupported values



FINANCIAL VALIDATION

Balance Sheet

- $\text{Assets} = \text{Liabilities} + \text{Equity}$

Profit & Loss

- $\text{Profit / Loss} = \text{Income} - \text{Expenses}$

Cash Flow

- $\text{Closing Cash} = \text{Opening Cash} + \text{Net Cash Flow}$

Invoice

- $\text{Total} = \text{Subtotal} + \text{Tax} - \text{Discounts}$



FINANCIAL VALIDATION

Validation Output

- Formula / Check
- Inputs
- Calculated Value
- Reported Value
- Variance
- PASS / FAIL / NOT_APPLICABLE



REST API & PERSISTENCE

Core Endpoints

- POST /api/v1/documents/process
- GET /api/v1/documents/{document_name}
- GET /api/v1/documents
- GET /api/v1/health

Processing

- Multipart file upload
- Document type supplied by frontend
- Synchronous processing
- Latest result persisted by filename



REST API & PERSISTENCE

Persistence

- PostgreSQL
- Local SQLite fallback for development
- Processed documents remain available through the dashboard

API Documentation

- OpenAPI / Swagger UI
 - `/docs`



FRONTEND DASHBOARD

 **NeoStats**
Document Intelligence

Dashboard

+ Process Document

DOCUMENTS

All Documents

API Connected

v1.0.0

Document *Intelligence*

Turn your financial documents into structured data with AI.

Financial Document **Processing**

Upload a financial document and extract structured information with automated validation.

Invoices

Balance Sheets

Profit & Loss

Cash Flow Statements

Extract

Validate

Analyze

Gain Insights

Process Document

Select the document type and upload a PDF or image.

Document Type

Invoice

Select Document

Choose File

20251118_0...2_11zon.jpg

Supported: PDF, JPG, PNG · Maximum 3 pages

20251118_000612_11zon.jpg

Ready to process

981.0 KB

Process Document

FRONTEND DASHBOARD

Dashboard Features

- Document type selection
- PDF / JPG / PNG upload
- Process Document
- API connection status
- Processed document list
- Search processed documents
- View previous results
- Extracted fields & line items
- Validation results
- Missing / failed value highlighting
- Raw JSON output
- Copy JSON

Designed for quick review of extracted financial information and validation results.



TESTING & DEPLOYMENT

Automated tests

- API endpoint tests
- Financial validation tests
- Validation service tests
- Invoice GST calculation tests
- Multiple financial statement scenarios

Deployment

Component	Technology
Frontend	Render Static Site
Backend	Render
Database	PostgreSQL
OCR	OCR.space
AI	Gemini 3.6 Flash



AI USAGE DECLARATION

- ChatGPT was used as an AI coding assistant for architecture guidance, debugging, testing and implementation support.
- Gemini is used as the document extraction model in the application.
- AI-assisted code was reviewed, tested and modified during implementation.
- API credentials are loaded from environment variables and are not committed to source control.



THANKYOU

Presented by SIVA D M

